INCH-POUND
MIL-DTL-2781H
12 January 2023
SUPERSEDING
MIL-DTL-2781G
11 August 2020

DETAIL SPECIFICATION

INSULATION, PIPE, THERMAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers preformed thermal insulation for use on pipes at surface temperatures up to 1,200 degrees Fahrenheit (°F).

1.2 Classification. Thermal pipe insulation is of the following types, as specified (see 6.2).

a. Type I – Calcium silicate high temperature insulation

b. Type II – Compressed E-glass high temperature insulation

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-2861 - Cement, Insulation, High Temperature

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1623 - Fire Performance Requirements and Approved Specifications for Interior Finish Materials and Furnishings (Naval Shipboard Use)

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5640

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9510-AB-ATM-010 - Nuclear Powered Submarine Atmosphere Control Manual; Volume 1

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of the chapter of S9510-AB-ATM-010 titled "Material Control Program" are available by email request to CommandStandards@navy.mil. Copies of T9070-AL-DPC-020/077-2 are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM C165 - Standard Test Method for Measuring Compressive Properties of Thermal Insulations

ASTM C203 - Standard Test Methods for Breaking Load and Flexural Properties of Block-Type Thermal Insulation

ASTM C302 - Standard Test Method for Density and Dimensions of Preformed Pipe-Covering-Type Thermal Insulation

ASTM C335/C335M - Standard Test Method for Steady-State Heat Transfer Properties of Pipe Insulation

ASTM C356 - Standard Test Method for Linear Shrinkage of Preformed High-Temperature Thermal Insulation Subjected to Soaking Heat

ASTM C411 - Standard Test Method for Hot-Surface Performance of High-Temperature Thermal Insulation

ASTM C421 - Standard Test Method for Tumbling Friability of Preformed Block-Type and Preformed Pipe-Covering-Type Thermal Insulation

ASTM C449 - Standard Specification for Mineral Fiber Hydraulic-Setting Thermal Insulating and Finishing Cement

ASTM C547 - Standard Specification for Mineral Fiber Pipe Insulation

ASTM C585 - Standard Practice for Inner and Outer Diameters of Thermal Insulation for Nominal Sizes of Pipe and Tubing

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. The thermal pipe insulation furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Material. The thermal pipe insulation shall consist of heat-resisting compounds meeting the requirements for the temperature conditions and the purpose intended.

3.3 Prohibited materials. The thermal pipe insulation shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2. Neither asbestos nor ceramic (refractory) fibers nor materials containing any of these fibers shall be used in insulation.

3.4 Form and dimensions.

3.4.1 Form and length. Thermal pipe insulation in nominal pipe sizes up to and including 10 inches shall be furnished in sections either 3 feet or 3.28 feet long, split in half lengthwise. Pipe insulation for nominal pipe sizes larger than 10 inches may be furnished split in half or furnished in semi-cylindrical sections or as curved segments, 3 feet or 3.28 feet in length. Each section or segment shall be true to shape and roundness and shall fit standard iron pipe sizes. A tolerance of ±¼ inch in length will be permitted (see 4.4.2.1).

3.4.2 Inner diameter. The inner diameter of the pipe insulation shall be in accordance with ASTM C585 (see 4.4.2.2). The inner diameter of the pipe insulation listed in ASTM C585 approximates the outer diameter of iron pipe sizes. The insulation can be applied either to a pipe or as a second layer to a smaller size of pipe insulation.

3.4.3 Thickness. Pipe insulation shall be furnished in the thicknesses listed in ASTM C585, as specified (see 6.2). A tolerance of +⅛ inch or -1/16 inch in actual thicknesses in accordance with ASTM C585 will be permitted.

3.5 Physical requirements. Pipe insulation shall conform to the requirements specified in table I.

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TABLE I. Physical requirements.

Characteristics	Test	Physical requirements (type I)	Physical requirements (type II)
Density average, lb/ft^3, maximum	4.4.3	15.0	12.5
Thermal conductivity (average), Btu-in/hr. ft^2 °F at mean temperature of:	4.4.4		
200 °F		0.42	0.35
300 °F		0.45	0.40
400 °F		0.50	0.44
500 °F		0.60	0.50
600 °F		0.65	0.56
700 °F		0.70	0.64
Weight loss by tumbling, loss in weight (average) percent, maximum	4.4.5		
After first 10 minutes		20	10
After second 10 minutes		40	20
Changes under soaking heat for 6 hours at 1,200 °F, linear shrinkage (average) percent, maximum	4.4.6	2.0	2.0
Compressive strength at not more than 5 percent deformation, lb/in^2, minimum	4.4.10	60	5
Compression resiliency, percent, minimum	4.4.12	N/A	90
Flexural strength, lb/in^2, minimum	4.4.11	1/	1/

FOOTNOTE:

1/ Requirement calculated by multiplying the density (as reported in lb/ft^3) by 3 and reporting in lb/in^2.

NOTE: Abbreviations not specified elsewhere herein include pound per cubic foot (lb/ft^3), British thermal unit inch per hour per square foot (Btu-in/hr. ft^2), and pound per square inch (lb/in^2).

3.6 Simulative service. Thermal pipe insulation shall not warp, crack, or show other visible changes. Minor surface cracks that do not propagate more than ¼ inch internally from the hot-face surface of the pipe insulation shall be disregarded (see 4.4.8).

3.7 Flame spread and smoke density. The finished insulation shall conform to the flame spread and smoke density requirements set forth in MIL-STD-1623 (see 4.4.7).

3.8 Compatibility. The insulation material shall be compatible with thermal insulation cement in accordance with ASTM C449 and with high temperature thermal insulating cement in accordance with MIL-C-2861. Without the application of a primer, the cements shall readily adhere to the insulation segments and form a smooth, protective surface. Drying time for each cement after application shall not exceed 2 hours (see 4.4.9).

3.9 Appearance and workmanship. Insulation shall be free of defects in appearance and workmanship (see 4.4.2.4). Since some requirements of workmanship for this material are not easily defined by a numerical value, the insulation shall not have visible defects that will adversely affect its service qualities.

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3.10 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.11 Toxicity. When evaluated in accordance with 4.5, the thermal pipe insulation shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.5 and 6.5).

3.12 Off-gassing. The thermal pipe insulation shall be evaluated for off-gassing in accordance with the requirements of 4.7. Based on the circumstances of use and the chemical nature of the thermal pipe insulation, the Navy will determine whether off-gas testing is required or if an administrative assessment is acceptable. In order to be considered acceptable for use in submarines, the thermal pipe insulation shall be assigned to either the "Permitted" or "Limited" category (see 4.7 and 6.7).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.1.1 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.4.

4.2 Qualification inspection. Qualification inspection shall include the examinations and tests listed in table II. Qualification tests shall be conducted at a laboratory acceptable to NAVSEA.

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TABLE II. Qualification and conformance inspections.

Inspection	Qualification	Conformance	Requirement	Test method
Material	X	X	3.2	Certification and visual
Prohibited materials	X		3.3	4.6
Dimensions	X	X	3.4	4.4.2
Density	X	X	3.5	4.4.3
Thermal conductivity	X		3.5	4.4.4
Weight loss by tumbling	X		3.5	4.4.5
Changes under soaking heat (6 hours at 1,200 °F), linear shrinkage	X		3.5	4.4.6
Flame spread and smoke density	X		3.7	4.4.7
Simulative performance	X		3.6	4.4.8
Compatibility	X		3.8	4.4.9
Compressive strength at 5 percent deformation	X	X	3.5	4.4.10
Compression resiliency 1/	X	X	3.5	4.4.12
Flexural strength	X	X	3.5	4.4.11
Appearance and workmanship	X	X	3.9	4.4.2.4
Toxicity	X		3.11	4.5
Off-gassing	X		3.12	4.7

NOTE:
1/ Not applicable for type I.

4.2.1 Qualification sample. Three samples for qualification tests shall be selected and subjected to the tests of 4.4.2, 4.4.3, 4.4.5, and 4.4.6. One sample shall be selected for the tests of 4.4.4, 4.4.7, 4.4.8, and 4.4.9.

4.3 Conformance inspection. Conformance inspection shall include the examinations and tests listed in table II.

4.3.1 Lot. For the purpose of this specification, a lot shall consist of one production batch produced at one time, of the same type, under the same conditions, and from ingredients of the same kind, nature, and quality. However, for molded thermal insulation that is produced by a continuous process, a lot shall consist of all insulation produced under the same conditions and from ingredients of the same kind, nature, and quality offered for delivery at one time.

4.3.2 Sampling for conformance inspection. Regarding a lot produced in a batch, a sample quantity of insulation sections from each lot shall be selected in accordance with table III and subjected to the examinations and tests listed in table II. For molded thermal pipe and block insulation that is produced by a continuous process, sampling shall be in accordance with table IV and subjected to the examinations and tests listed in table II. If one or more defects are found in any sample, the entire lot shall be rejected. The contractor has the option of screening 100 percent of the lot for the defective characteristics or providing a new lot, which shall be inspected in accordance with the sampling plan herein.

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TABLE III. Batch lot sampling for conformance inspection.

Lot size	Minimum sample size
2 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
501 to 1,200	15
1,201 to 3,200	18
3,201 to 10,000	22
10,001 or greater	29

TABLE IV. Continuous lot sampling for conformance inspection.

Lot size of pieces [1/]	Minimum sample size
Up to 63	1
64 to 160	2
161 to 400	3
401 to 1,000	5
1,001 to 2,500	8
2,501 to 6,300	13
6,301 to 16,000	20
16,001 or greater	32
NOTE: [1/] A piece is defined as one segment, section, or block.	

4.4 Tests.

4.4.1 Conditioning samples. Test specimens shall be conditioned by drying to constant weight in an oven at a temperature of 215 to 250 °F.

4.4.2 Dimensions.

4.4.2.1 Length. Length shall be determined in accordance with ASTM C302 (see 3.4.1).

4.4.2.2 Diameter. The inner diameter of the pipe insulation shall be determined in accordance with ASTM C585 (see 3.4.2).

4.4.2.3 Thickness. Thickness shall be determined in accordance with ASTM C302 (see 3.4.3).

4.4.2.4 Appearance and workmanship. The pipe insulation shall be examined to determine that none of the following defects exist that will adversely affect its service qualities (see 3.9):

a. Cracked or broken lengths.

b. Friable or lumpy surface.

c. Inner core of insulation not concentric with outer cylindrical surface.

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d. Edges not plane.

e. Longitudinal cut surfaces not plane.

f. Excessive voids.

g. Warping.

h. Form not as required.

4.4.3 Density. The density shall be determined in accordance with ASTM C302 (see table I).

4.4.4 Thermal conductivity. Thermal conductivity shall be determined in accordance with ASTM C335/C335M (see table I).

4.4.5 Weight loss by tumbling. Weight loss by tumbling shall be determined in accordance with ASTM C421 (see table I).

4.4.6 Physical changes after heat soaking. Specimens shall be weighed and measured. The specimens shall then be placed in an electrically heated oven and subjected to 1,200 °F for 6 hours. The specimens shall then be removed from the oven, cooled to room temperature, and tested to determine linear shrinkage in accordance with ASTM C356.

4.4.7 Flame spread and smoke density. Specimens shall be tested in accordance with MIL-STD-1623.

4.4.8 Simulative performance. Simulative performance shall be determined in accordance with ASTM C411 (see 3.6). Exposure temperature shall be 1,200 °F.

4.4.9 Compatibility. A 6-inch section of pipe insulation, 1½ inches thick, shall be uniformly surface-coated while at room temperature with an approximately ¼-inch layer of finishing cement in accordance with ASTM C449. Drying time under ambient air temperature conditions shall be determined. This test shall be repeated with a second pipe section using a cement in accordance with MIL-C-2861. The drying time for this cement shall also be determined.

4.4.10 Compressive strength. The compressive strength shall be determined in accordance with ASTM C165. A block sample may be used, provided it is from the same lot as the pipe sections.

4.4.11 Flexural strength. The flexural strength shall be determined in accordance with ASTM C203. A block sample may be used, provided it is from the same lot as the pipe sections.

4.4.12 Compression resiliency. The compression resiliency shall be determined in accordance with ASTM C547. A block sample may be used, provided it is from the same lot as the pipe sections.

4.5 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.11 as required by the qualifying activity. The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the thermal pipe insulation using data provided by the manufacturer/distributor to the NMCPHC (see 3.11 and 6.5).

4.6 Prohibited materials. Prohibited materials shall be verified for conformance to 3.3 as required by the qualifying activity.

4.7 Off-gassing. The thermal pipe insulation shall be evaluated for off-gassing in accordance with S9510-AB-ATM-010 chapter titled “Material Control Program” (see 3.12 and 6.7). If the Navy determines that off-gas testing is required, testing shall be conducted at a NAVSEA approved test facility (see 3.12). The Navy will review the off-gas test results and assign a usage category. Additionally, the Navy will assign a usage category if an administrative review is conducted in lieu of off-gas testing (see 3.12).

4.8 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is acceptable to NAVSEA. Accreditation shall be obtained from a recognized accreditation body such as American Association for Laboratory Accreditation (A2LA) (www.A2LA.org) or International Code Council’s International Accreditation Services (IAS) (www.IASonline.org). The scope of accreditation shall include specific flammability and fire tests required for qualification. All other fire test provisions shall be as specified (see 6.2 and 6.8).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The thermal insulation is intended to provide for piping operating at surface temperatures up to 1,200 °F in U.S. Navy Ships.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type (see 1.2).

c. Nominal iron pipe size and nominal thickness (see 3.4.3).

d. Inspection conditions, if different (see 4.1.1).

e. Additional fire testing provisions (see 4.8 and 6.8).

f. Packaging requirements (see 5.1).

g. When safety data sheets (SDS) are required (see 6.4).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 2781 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Safety data sheets. When specified (see 6.2), contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. In order to obtain an SDS, Federal Acquisition Regulation (FAR) clause 52.223-3 must be in the contract.

6.5 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, https://www.med.navy.mil/Navy-Marine-Corps-Public-Health-Center/.

6.6 Certification. Consideration should be given to including certificates of compliance with each shipment of insulation. Certificates should indicate successful completion of the individual tests of the conformance inspection.

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6.7 Material certification. Materials to be installed in submarines are to be controlled to prevent off-gassing, which contaminates the submarine's atmosphere and can result in health hazards to personnel or deleterious effects on machinery. These controls are administered through the Submarine Material Control Program, which is described in the Nuclear Powered Submarine Atmosphere Control Manual, S9510-AB-ATM-010 chapter titled "Material Control Program." Under the Submarine Material Control Program, all materials considered for use on submarines require certification and assignment of a usage category. Under the certification process, candidate materials are selected by Navy activities or contractors, and a request for certification is submitted to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. The certification request is accompanied by detailed information, including descriptions of the material, method of application, usage, and storage. A chemical analysis is conducted, which can be accomplished through off-gas testing. If off-gas testing is required, it must be conducted in a Government approved laboratory. Information pertaining to this test requirement may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. Based on the chemical analysis results, a usage category is assigned to the material defining whether, and to what extent, the material may be used on submarines.

6.8 Additional fire testing provisions. NAVSEA reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements of the qualification tests. A2LA (www.A2LA.org) or International Code Council's IAS (www.IASonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

6.9 Subject term (key word) listing.

Flame spread

Smoke density

Thermal conductivity

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH

Preparing activity:
Navy – SH
(Project 5640-2022-003)

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